THE FUNDAMENTAL DISTINGUISHED TRIANGLE FOR A COMPOSITION OF RING MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the triangle; the proof derives it from the short exact sequence of sheaves and then performs the three identifications.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The short exact sequence of differential sheaves on the triple category. Lemma lemma-triangle-ses (1389-1410, produced as L44139), used at proof lines 1597-1601. Is the sequence to which the derived functor is applied.
- The computation of the lower shrieks along the three projections. Lemma lemma-triangle-compute-lower-shriek (1506-1512, produced as L44131), used at proof lines 1608-1612. Identifies the three terms of the triangle.
- The computation of the cotangent complex from the differential sheaves. Lemma lemma-compute-cotangent-complex, used at proof lines 1608-1616. Matches the lower shrieks with the cotangent complexes.
- The change-of-rings lemma and flatness of the sheaf of differentials. Cohomology on Sites, Lemma sites-cohomology-lemma-change-of-rings, used at proof lines 1617-1620. Rewrites the first term as a derived tensor product.

Source reference. Original lemma 08PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11037–11037. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11074–11074. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11072–11072. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11051–11051. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Proposition 0.2. *Let $A \to B \to C$ be ring maps. There is a canonical distinguished triangle*

$$L_{B/A} \otimes^{\mathbf{L}}_B C \to L_{C/A} \to L_{C/B} \to L_{B/A} \otimes^{\mathbf{L}}_B C[1]$$

in $D(C)$.

Proof. Consider the short exact sequence of sheaves of Lemma 08QU and apply the derived functor $L\pi_!$ to obtain a distinguished triangle

$$L\pi_!(g_1^{-1}\Omega_1 \otimes_{\underline{B}} \underline{C}) \to L\pi_!(g_2^{-1}\Omega_2) \to L\pi_!(g_3^{-1}\Omega_3) \to L\pi_!(g_1^{-1}\Omega_1 \otimes_{\underline{B}} \underline{C})[1]$$

in $D(C)$. Using Lemmas 08QW and 08PU we see that the second and third terms agree with $L_{C/A}$ and $L_{C/B}$ and the first one equals

$$L\pi_{1,!}(\Omega_1 \otimes_{\underline{B}} \underline{C}) = L\pi_{1,!}(\Omega_1) \otimes^{\mathbf{L}}_B C = L_{B/A} \otimes^{\mathbf{L}}_B C$$

The first equality by Cohomology on Sites, Lemma 08Q8 (and flatness of Ω_1 as a sheaf of modules over $\underline{B}$) and the second by Lemma 08PU. □